Advanced Mathematical Techniques (AMT) Notes

1 Today's stuff

Recall that there are 2 types of asymptotic expansions - one that we integrate by parts, and the other that has a suppressed value for a long time so we simplify the integral. Note that these tend to be alternating series. Looking at an example with integration by parts, if we let:

$$f(x) = e^x \int_x^\infty \frac{e^{-t}}{\sqrt[3]{t}} dt = e^x(-t^{-1/3}e^{-t} + \frac{1}{3}t^{-4/3}e^{-t} - \frac{4}{9}t^{-7/3}e^{-t})\Big|_x^\infty - \frac{e^x \cdot 28}{27} \int_x^\infty t^{-10/3}e^{-t} dt$$